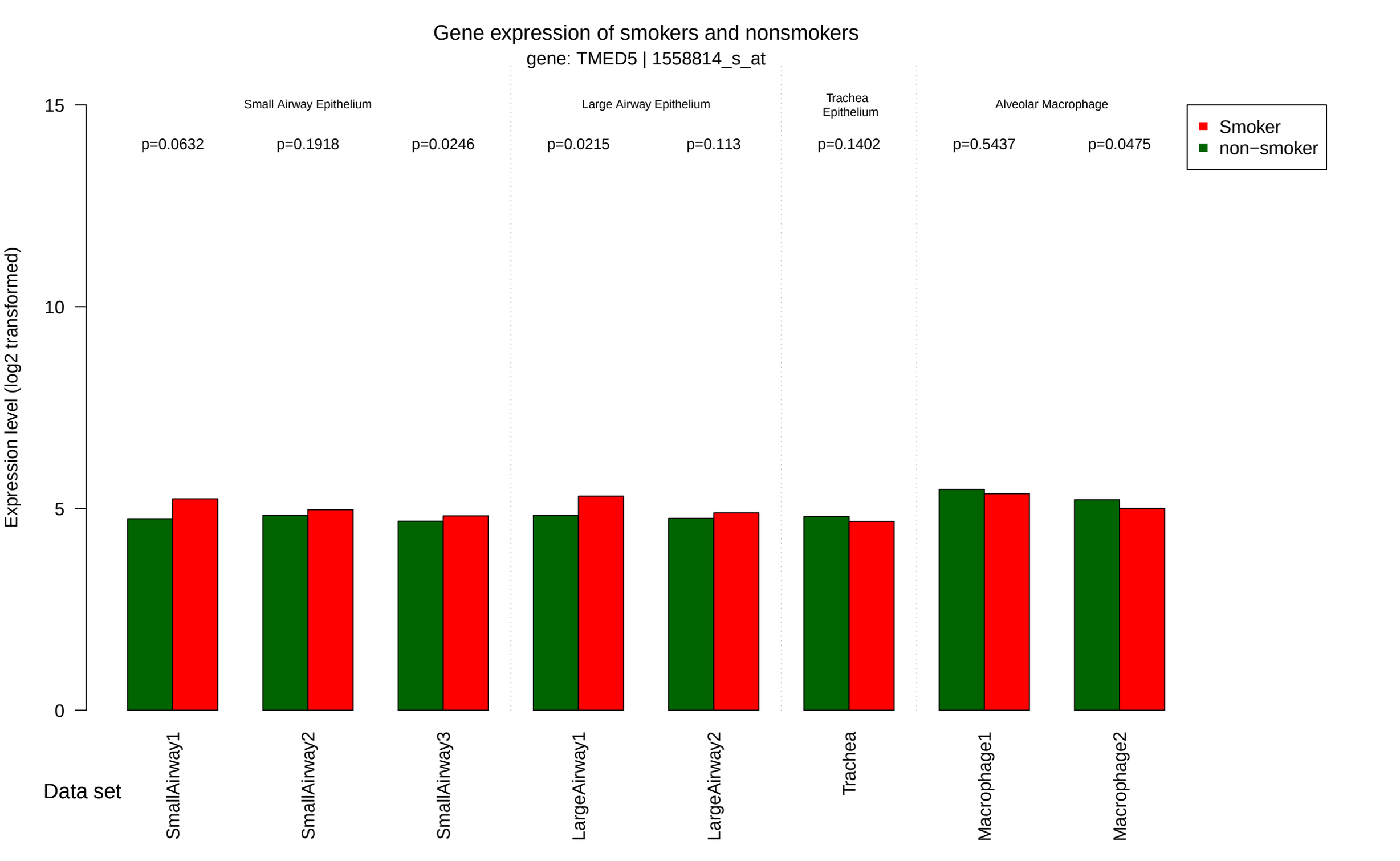
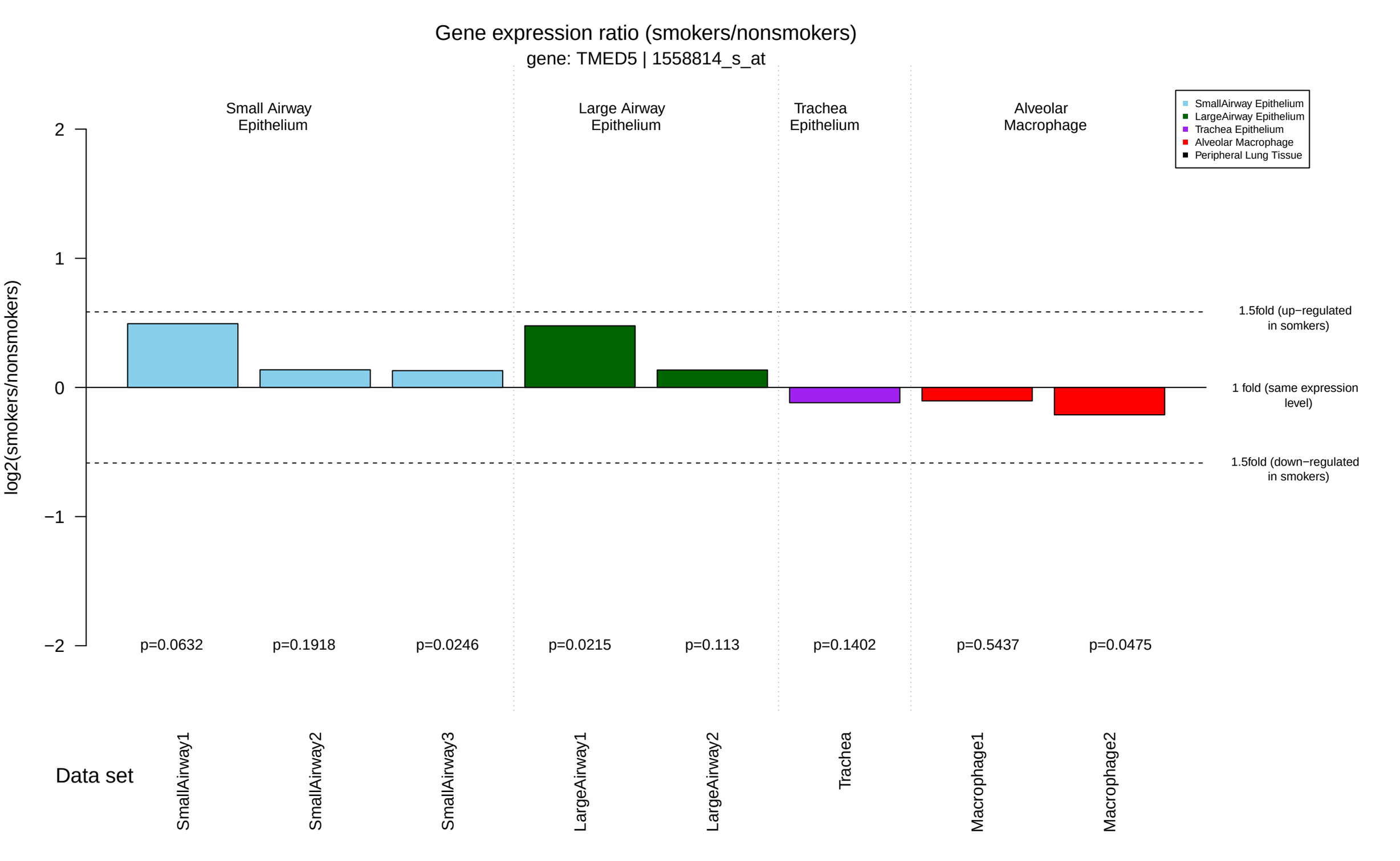
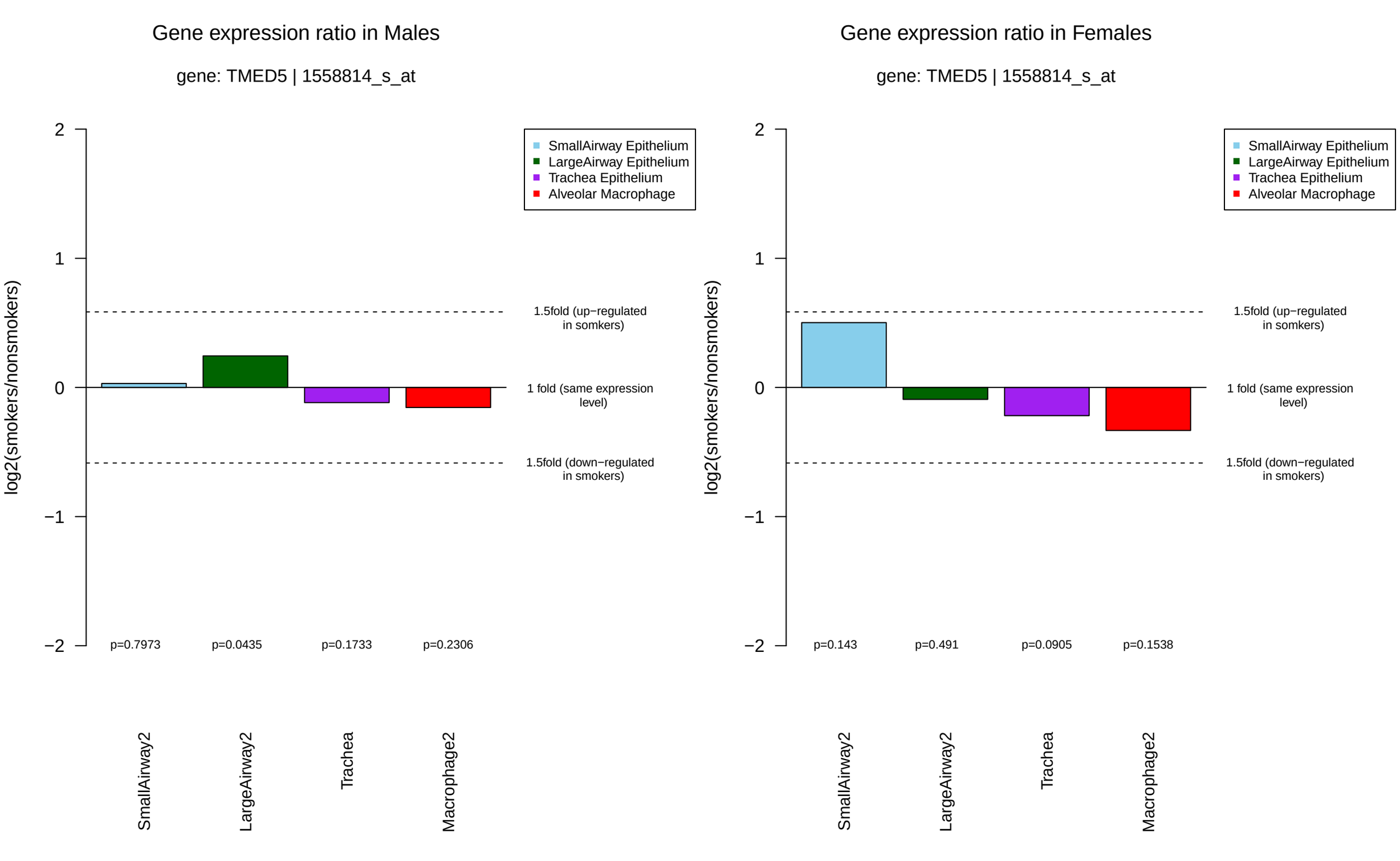
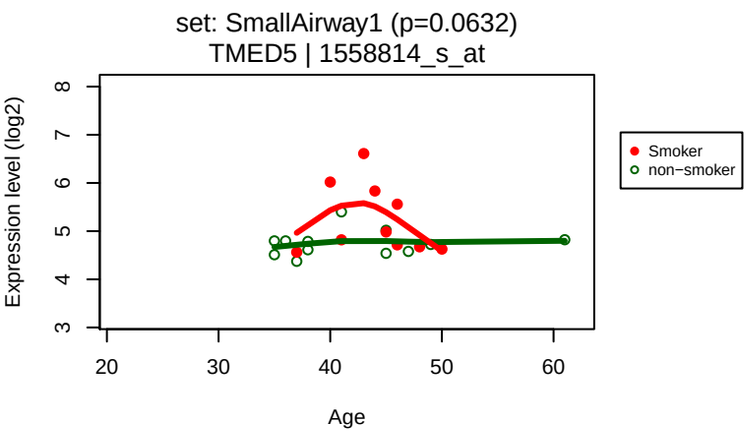
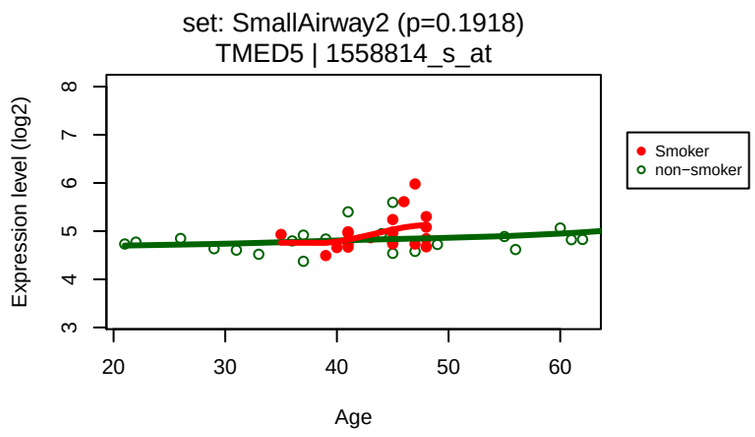
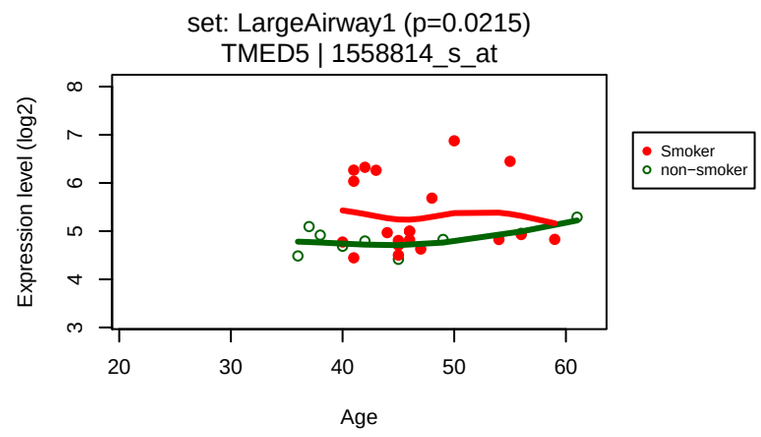

Gene expression of smokers and nonsmokers
gene: TMED5 | 1558814_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0632
p=0.1918
p=0.0246
p=0.0215
p=0.113
p=0.1402
p=0.5437
p=0.0475
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

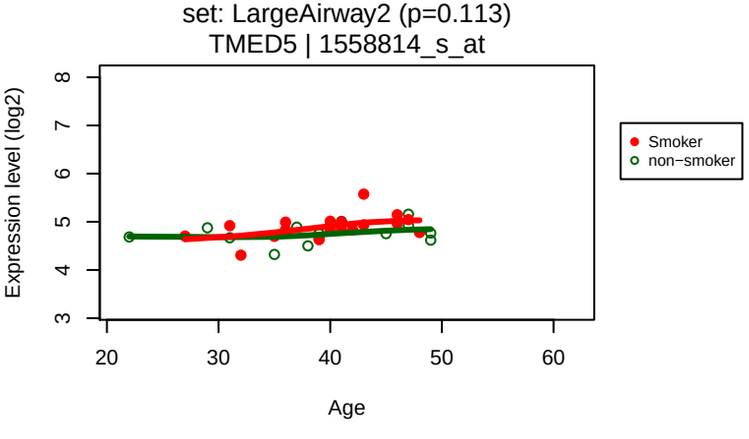

Gene expression ratio (smokers/nonsmokers)
gene: TMED5 | 1558814_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0632
p=0.1918
p=0.0246
p=0.0215
p=0.113
p=0.1402
p=0.5437
p=0.0475
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

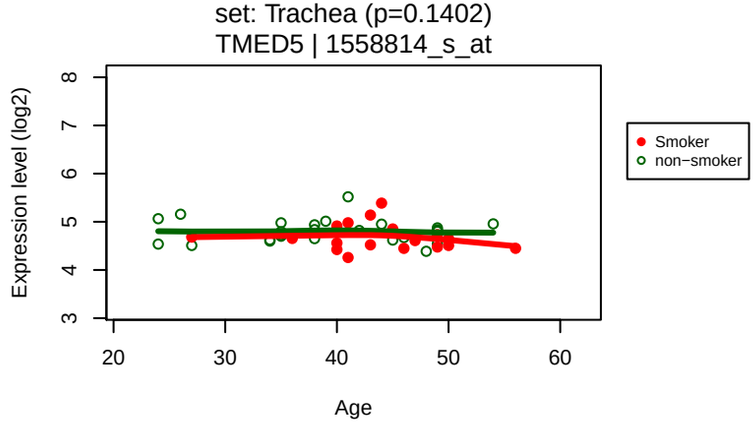

Gene expression ratio in Males
gene: TMED5 | 1558814_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7973
p=0.0435
p=0.1733
p=0.2306
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMED5 | 1558814_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.143
p=0.491
p=0.0905
p=0.1538
SmallAirway2
LargeAirway2
Trachea
Macrophage2

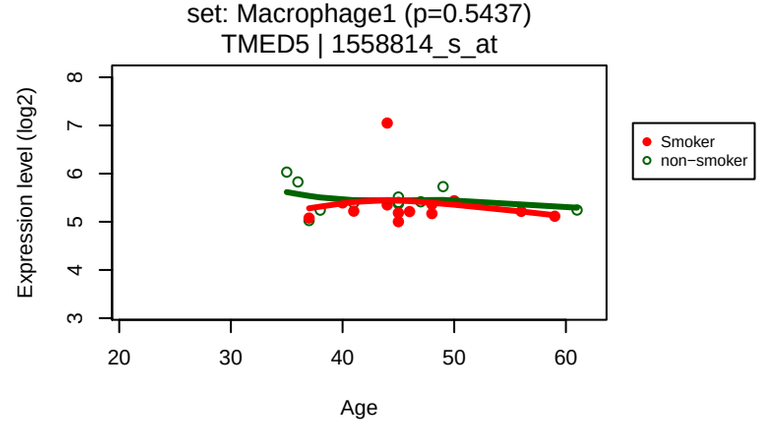

set: SmallAirway1 (p=0.0632)
TMED5 | 1558814_s_at
Expression level (log2)
Age
Smoker
non−smoker

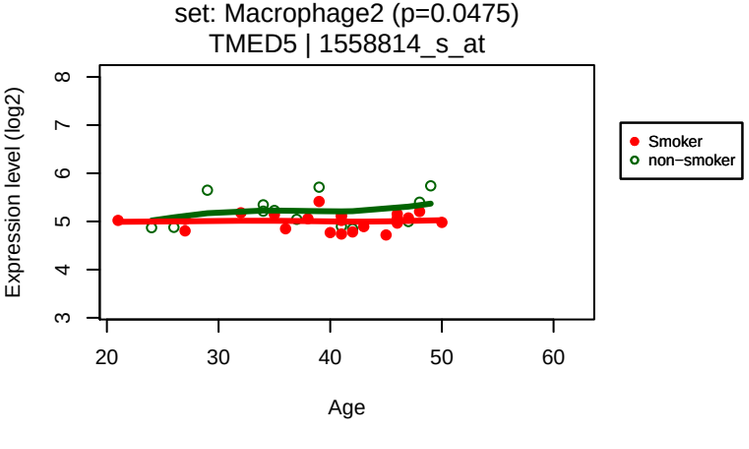

set: SmallAirway2 (p=0.1918)
TMED5 | 1558814_s_at
Expression level (log2)
Age
Smoker
non−smoker

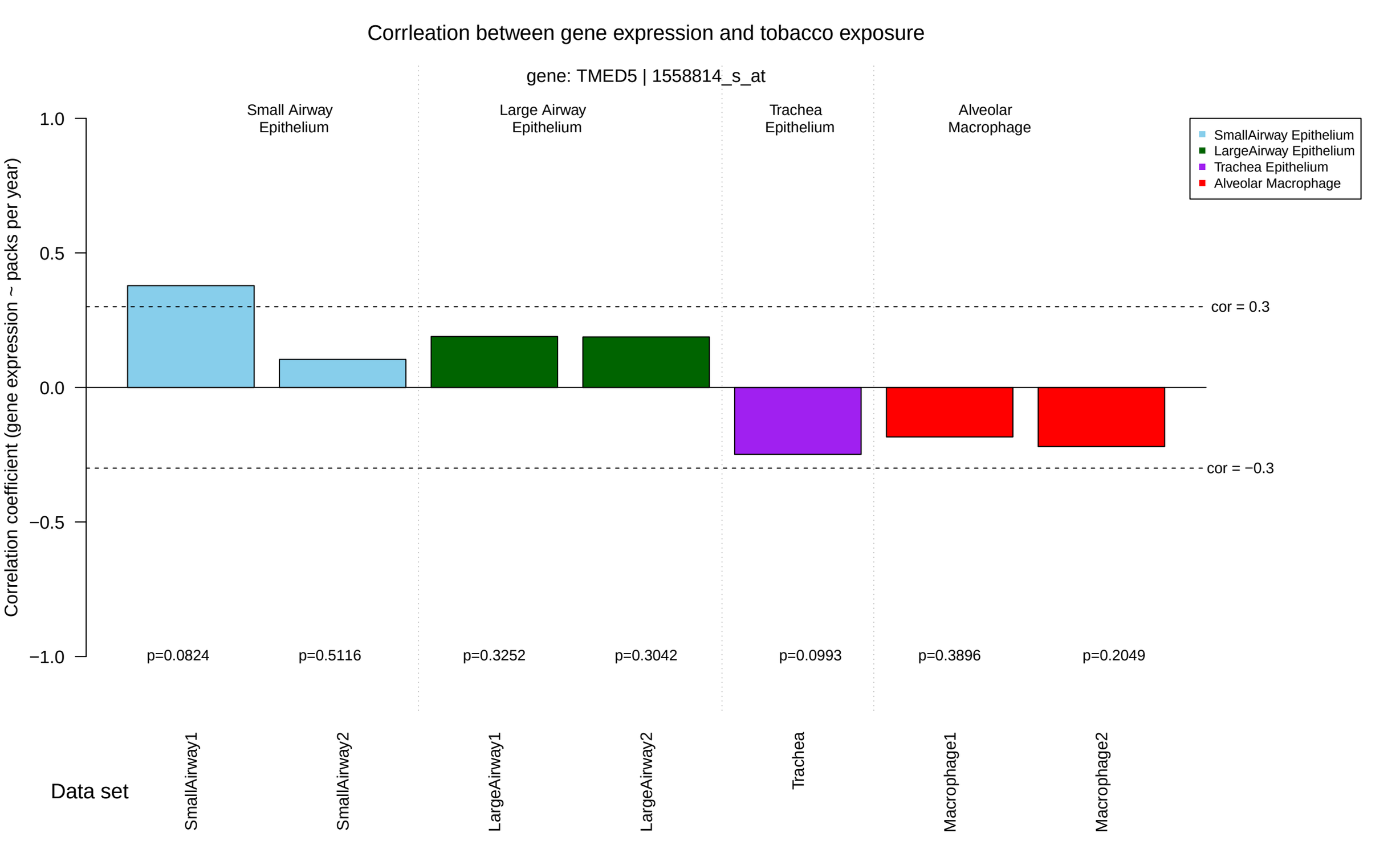

set: LargeAirway1 (p=0.0215)
TMED5 | 1558814_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.113)
TMED5 | 1558814_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1402)
TMED5 | 1558814_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5437)
TMED5 | 1558814_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0475)
TMED5 | 1558814_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMED5 | 1558814_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0824
p=0.5116
p=0.3252
p=0.3042
p=0.0993
p=0.3896
p=0.2049
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2